Chapter 5

Quantum dynamics and quantum computing

The quantum technologies and quantum computers currently developed in academia and in a number of commercial companies are based on how quantum systems interact with each other and with external electromagnetic fields. Much of the underlying dynamics can be understood from simple models involving a small number of states. This project will explore this dynamics, starting with a short study of the entanglement of two atomic qubits.

5.1 Introduction

A qubit is a quantum system which, for all practical purposes, can be considered to be a two-state system. I.e., a qubit can be in a state $|0\rangle$, or in a state $|1\rangle$ orthogonal to $|0\rangle$, or in a state described by a linear combination of $|0\rangle$ and $|1\rangle$, and cannot be in a state orthogonal to both $|0\rangle$ and $|1\rangle$. Many different systems meet this requirement. E.g., $|0\rangle$ and $|1\rangle$ could be two energy eigenstates of a certain atom, in a situation where there is effectively zero probability that the atom could make a transition to an energy eigenstate other than $|0\rangle$ and $|1\rangle$. However, certain quantum states of light, various condensed matter systems and some nuclear spin states can also be used to form qubits.

The state vector describing the quantum state of a qubit can be written as

$$|\psi\rangle \quad = \quad \alpha|0\rangle + \beta|1\rangle$$

where α and β are two complex numbers. The vectors $|0\rangle$ and $|1\rangle$ act as basis vectors here: the set $\{|0\rangle, |1\rangle\}$ is the basis in which we work. As already mentioned, we assume that $|0\rangle$ and $|1\rangle$ are orthogonal: $\langle 0|1\rangle = \langle 1|0\rangle = 0$. We also assume that these two vectors are normalised: $\langle 0|0\rangle = \langle 1|1\rangle = 1$. Equivalently, we can write $|\psi\rangle$ in the form a two-component column vector as

$$|\psi\rangle \quad = \quad \begin{pmatrix} \alpha \\ \beta \end{pmatrix}.$$

In this representation, the Hamiltonian acting on that state vector would be a 2×2 matrix, rather than a certain operator acting on the vectors $|0\rangle$ and $|1\rangle$.

Quantum computing is based on making several qubits interact with each other in a useful way. Instead of looking at these qubits separately, as as many independent systems, one considers the system formed

by all the qubits of interest, taken together as a single system. Through their interaction, the quantum state of these qubits may become correlated to each other, so that the whole system is in an entangled state (see below). The art is to produce the desired entangled state and manipulate it as required to do the intended computation.

Consider, for example, a system formed by two different qubits, namely a qubit A and a qubit B (e.g., two separated atoms, the two atoms being considered as forming a single system). The basis states for that system can be taken to be $|00\rangle$, $|01\rangle$, $|10\rangle$ and $|11\rangle$, describing, respectively, a state of the two-qubit system in which qubits A and B are both in state $|0\rangle$, a state in which qubit A is in state $|0\rangle$ and qubit B in state $|1\rangle$, etc. A general state of this two-qubit system can be written as a linear combination of these four basis vectors, e.g.,

$$|\Psi\rangle \;=\; a_{00}|00\rangle + a_{01}|01\rangle + a_{10}|10\rangle + a_{11}|11\rangle \;,$$

where a_{00}, a_{01}, a_{10} and a_{11} are four complex numbers. Alternatively, we can write

$$|\Psi\rangle \;=\; \begin{pmatrix} a_{00} \\ a_{01} \\ a_{10} \\ a_{11} \end{pmatrix} .$$

Any operator acting on this state, such as the Hamiltonian, will be a 4×4 matrix.

What one means by entangled state here can be stated simply: it is a state which is not a simple product of a state of qubit A and a state of qubit B. For an example of a product state, assume, e.g., that qubit A is in the state $\alpha_A|0\rangle_A + \beta_A|1\rangle_A$ and qubit B in the state $\alpha_B|0\rangle_B + \beta_B|1\rangle_B$. Then the joint state of these two qubits would be

$$\alpha_A\alpha_B|00\rangle + \alpha_A\beta_B|01\rangle + \beta_A\alpha_B|10\rangle + \beta_A\beta_B|11\rangle,$$

i.e., the state $|\Psi\rangle$ with $a_{00} = \alpha_A\alpha_B$, $a_{01} = \alpha_A\beta_B$, $a_{10} = \beta_A\alpha_B$ and $a_{11} = \beta_A\beta_B$. When the 2-qubit system is in such a product state, the state of qubit A is effectively independent of the state of qubit B. Contrast this with, e.g., the state

$$|\Psi^+\rangle \;=\; \frac{1}{\sqrt{2}}|01\rangle + \frac{1}{\sqrt{2}}|10\rangle \;,$$

i.e., $a_{00} = 0$, $a_{01} = 1/\sqrt{2}$, $a_{10} = 1/\sqrt{2}$ and $a_{11} = 0$. There are no values of α_A, β_A, α_B and β_B giving these values of a_{00}, a_{01}, a_{10} and a_{11}. When in that joint state, the system is therefore not in a product state: it is in an entangled state. In such a state, the state of qubit A is related to the state of qubit B. (In fact, $|\Psi^+\rangle$ is one of the four maximally entangled Bell's states often considered in studies of quantum entanglement.)

The aim of the milestone is to model the dynamics of two interacting qubits driven by an external EM field (e.g., a laser field). We shall see how interactions give rise to quantum entanglement. First, we need to derive a Hamiltonian describing the interaction between a single qubit and an EM field.

5.2 Background theory

5.2.1 Derivation of the single-qubit Hamiltonian

We first look at how the state of a single qubit may evolve under the effect of some time-dependent interaction. The Schrödinger equation describing how the state of this qubit evolves in time is

$$\mathrm{i}\hbar\partial_t|\psi\rangle = (\mathcal{H}_0 + \mathcal{H}')|\psi\rangle \;, \tag{5.1}$$

where $\mathcal{H}_0$ is the Hamiltonian of the unperturbed qubit and $\mathcal{H}'$ is the part of the total Hamiltonian arising from the interaction ($\mathcal{H}'$ would typically be a potential energy term). We assume that the basis states $|0\rangle$ and $|1\rangle$ are eigenstates of $\mathcal{H}_0$ and denote the corresponding eigenenergies by E_0 and E_1:

$$\mathcal{H}_0|0\rangle = E_0|0\rangle \;, \qquad \mathcal{H}_0|1\rangle = E_1|1\rangle \;.$$

In the absence of the interaction, the Schrödinger equation would simply be

$$\mathrm{i}\hbar\partial_t|\psi\rangle = \mathcal{H}_0|\psi\rangle \;,$$

whose solution would have the general form

$$|\psi(t)\rangle \;=\; a|0\rangle\mathrm{e}^{-\mathrm{i}E_0t/\hbar} + b|1\rangle\mathrm{e}^{-\mathrm{i}E_1t/\hbar} \tag{5.2}$$

where a and b are two constants. (Cf. the Foundations 2A QM course. Here we assume that $\mathcal{H}_0$ does not depend on time, which is generally the case in applications of this theory.) Solutions of this form are not possible when the interaction is present; instead, we seek a solution of Eq. (5.1) of the form

$$|\psi(t)\rangle \;=\; a(t)|0\rangle\mathrm{e}^{-\mathrm{i}E_0t/\hbar} + b(t)|1\rangle\mathrm{e}^{-\mathrm{i}E_1t/\hbar} \;,$$

where the coefficients a and b now vary in time. We substitute this state vector in Eq. (5.1) and cancel some terms on the left and right, leaving

$$\mathrm{i}\hbar\left(\dot{a}(t)|0\rangle\mathrm{e}^{-\mathrm{i}E_0t/\hbar} + \dot{b}(t)|1\rangle\mathrm{e}^{-\mathrm{i}E_1t/\hbar}\right) = \mathcal{H}'|0\rangle a(t)\mathrm{e}^{-\mathrm{i}E_0t/\hbar} + \mathcal{H}'|1\rangle b(t)\mathrm{e}^{-\mathrm{i}E_1t/\hbar} \;.$$

Next, we take the inner product with $\langle 0|\mathrm{e}^{\mathrm{i}E_0t/\hbar}$ or with $\langle 1|\mathrm{e}^{\mathrm{i}E_1t/\hbar}$. For the milestone, and in many other cases, the interaction of interest couples state $|0\rangle$ to state $|1\rangle$ but does not couple state $|0\rangle$ to itself or state $|1\rangle$ to itself. I.e., $\langle 0|\mathcal{H}'|0\rangle = \langle 1|H'|1\rangle = 0$. Using this fact, we find

$$\begin{aligned} \mathrm{i}\hbar\dot{a}(t) &= \langle 0|\mathcal{H}'|1\rangle\mathrm{e}^{-\mathrm{i}\omega_0t}\; b(t) \;, \\ \mathrm{i}\hbar\dot{b}(t) &= \langle 1|\mathcal{H}'|0\rangle\mathrm{e}^{\mathrm{i}\omega_0t}\; a(t) \;, \end{aligned}$$

with $\omega_0 = (E_1 - E_0)/\hbar$. We will assume that $E_1 > E_0$, so that $\omega_0 > 0$.

As already mentioned, the case we are interested in is an interaction of the qubit with an electromagnetic field. On general grounds, we can describe this interaction by an interaction Hamiltonian of the form $\mathcal{H}' = -\boldsymbol{d}\cdot\boldsymbol{\mathcal{E}}$, where $\boldsymbol{d}$ is the dipole operator and $\boldsymbol{\mathcal{E}}$ is the electric field component of the EM field. (Cf. the Foundations 3A QM3 course. Having studied that course, you'll recognize that we make the electric dipole approximation here.) We would simply have $\boldsymbol{d} = -e\boldsymbol{r}$ in the case of an atom of hydrogen, where e (positive) is the charge of the electron in absolute value; this expression generalises to much more

complicated cases. Moreover, we will assume that the wave is plane polarized (another term for this is linearly polarized) and write $\boldsymbol{\mathcal{E}} = \boldsymbol{\mathcal{E}}_0 \cos(\phi_{\rm L} - \omega t)$, where $\boldsymbol{\mathcal{E}}_0$ is a constant vector and $\phi_{\rm L}$ a constant phase-angle. We define the **Rabi frequency**, which is a measure of the coupling between the field and the qubit, as $\Omega = -\langle 1|\boldsymbol{d}\cdot\boldsymbol{\mathcal{E}}_0|0\rangle/\hbar$. In general $\langle 1|\boldsymbol{d}\cdot\boldsymbol{\mathcal{E}}_0|0\rangle$ is a complex number and $\langle 0|\boldsymbol{d}\cdot\boldsymbol{\mathcal{E}}_0|1\rangle$ is the complex conjugate of $\langle 1|\boldsymbol{d}\cdot\boldsymbol{\mathcal{E}}_0|0\rangle$; however, for simplicity we will assume that $\langle 1|\boldsymbol{d}\cdot\boldsymbol{\mathcal{E}}_0|0\rangle$ is real, so that $\langle 1|\boldsymbol{d}\cdot\boldsymbol{\mathcal{E}}_0|0\rangle = \langle 0|\boldsymbol{d}\cdot\boldsymbol{\mathcal{E}}_0|1\rangle$ and $\Omega = -\langle 0|\boldsymbol{d}\cdot\boldsymbol{\mathcal{E}}_0|1\rangle/\hbar$ as well. We can then rewrite the coupled equations as

$$\begin{aligned} \mathrm{i}\dot{a}(t) &= \Omega\cos(\phi_{\rm L}-\omega t)\ \mathrm{e}^{-\mathrm{i}\omega_0 t}\ b(t) \\ \mathrm{i}\dot{b}(t) &= \Omega\cos(\phi_{\rm L}-\omega t)\ \mathrm{e}^{\mathrm{i}\omega_0 t}\ a(t)\ . \end{aligned}$$

We now expand the cosine, using

$$\cos(\phi_{\rm L}-\omega t) = \tfrac{1}{2}\mathrm{e}^{\mathrm{i}\phi_{\rm L}}\mathrm{e}^{-\mathrm{i}\omega t} + \tfrac{1}{2}\mathrm{e}^{-\mathrm{i}\phi_{\rm L}}\mathrm{e}^{\mathrm{i}\omega t}.$$

Doing so gives terms in $\exp[\pm\mathrm{i}(\omega+\omega_0)t]$ and terms in $\exp[\pm\mathrm{i}(\omega-\omega_0)t]$. The case of interest is that where $\omega \approx \omega_0$, i.e., the case where the field is resonant or nearly resonant with the transition between states $|0\rangle$ and $|1\rangle$ (or in other words, the case where the photon energy, $\hbar\omega$, matches or nearly matches the transition energy $E_1 - E_0$). Typically $\omega+\omega_0 \sim 10^{15}\ \mathrm{s}^{-1}$ whereas $|\omega-\omega_0| \sim 10^7\ \mathrm{s}^{-1}$ or less in the applications we have in mind. The terms in $\exp[\pm\mathrm{i}(\omega+\omega_0)t]$ therefore oscillate much faster than the terms in $\exp[\pm\mathrm{i}(\omega-\omega_0)t]$ when t varies.The upshot is that we can neglect the former compared to the latter. Doing so is known as making the **rotating wave approximation**, for historical reasons. Making this approximation gives

$$\begin{aligned} \mathrm{i}\dot{a}(t) &= \tfrac{1}{2}\Omega\mathrm{e}^{-\mathrm{i}\phi_{\rm L}}\mathrm{e}^{\mathrm{i}\Delta t}\ b(t) \\ \mathrm{i}\dot{b}(t) &= \tfrac{1}{2}\Omega\mathrm{e}^{\mathrm{i}\phi_{\rm L}}\mathrm{e}^{-\mathrm{i}\Delta t}\ a(t)\ , \end{aligned}$$

where $\Delta = \omega - \omega_0$. ($\Delta$ is called the **detuning**.)

Finally, we transform into a so-called **rotating frame** by making the following change of variable:

$$\tilde{a}\,(t) = a(t)\,\mathrm{e}^{-\mathrm{i}(\Delta/2)t}, \qquad \tilde{b}(t) = b(t)\,\mathrm{e}^{\mathrm{i}(\Delta/2)t}\ .$$

This removes the explicit time dependence and we obtain

$$\begin{aligned} \mathrm{i}\dot{\tilde{a}} &= \tfrac{1}{2}\Delta\tilde{a} + \tfrac{1}{2}\Omega\mathrm{e}^{-\mathrm{i}\phi_{\rm L}}\ \tilde{b} \\ \mathrm{i}\dot{\tilde{b}} &= \tfrac{1}{2}\Omega\mathrm{e}^{\mathrm{i}\phi_{\rm L}}\ \tilde{a} - \tfrac{1}{2}\Delta\tilde{b}\ . \end{aligned}$$

We can rewrite these coupled equations in the form of a Schrödinger equation,

$$\mathrm{i}\hbar\partial t|\tilde{\psi}\rangle = \mathcal{H}_{\rm int}|\tilde{\psi}\rangle\ ,$$

with

$$\mathcal{H}_{\rm int} = \frac{\hbar}{2}\begin{pmatrix} \Delta & \Omega\mathrm{e}^{-\mathrm{i}\phi_{\rm L}} \\ \Omega\mathrm{e}^{\mathrm{i}\phi_{\rm L}} & -\Delta \end{pmatrix} \tag{5.3}$$

and

$$|\tilde{\psi}\rangle = \begin{pmatrix} \tilde{a} \\ \tilde{b} \end{pmatrix}\ .$$

We shall omit the tildes from now on. Eq. (5.3) is the single-qubit Hamiltonian that we shall use for our simultation. It depends on three parameters, namely the Rabi frequency, Ω, which is proportional to the electric field amplitude (and hence the square root of the intensity or power of the EM field), the detuning, Δ, which depends on the EM field frequency, and the phase of the field, $\phi_{\rm L}$.

5.2.2 The two-qubit Hamiltonian

Now we consider what happens when we apply the EM field described in the previous section to two qubits — i.e., each qubit evolves according to the Hamiltonian $\mathcal{H}_{\rm int}$ describing its interaction with the field. For clarity, we distinguish the Hamiltonian $\mathcal{H}_{{\rm int},A}$, which acts on the states of qubit A, from the Hamiltonian $\mathcal{H}_{{\rm int},B}$, which acts on the states of qubit B. We need to re-write these Hamiltonians in the two-qubit basis, so as to have the Hamiltonian acting on the two-qubit states. We can do this using tensor products, or in a more simple minded way by calculating each of the elements of the matrix representing the Hamiltonian in the $|00\rangle$, $|01\rangle$, $|10\rangle$ and $|11\rangle$ basis. For example,

$$\begin{aligned}\langle 00|\hat{H}|00\rangle &= {}_A\langle 0|\hat{H}_{{\rm int},A}|0\rangle_A \times {}_B\langle 0|0\rangle_B + {}_A\langle 0|0\rangle_A \times {}_B\langle 0|\hat{H}_{{\rm int},B}|0\rangle_B \\ &= \hbar\Delta/2 + \hbar\Delta/2.\end{aligned}$$

Proceeding in this way element by element yields the matrix representing the two-qubit Hamiltonian in that basis:

$$\mathcal{H} = \frac{\hbar}{2}\begin{pmatrix} 2\Delta & \Omega e^{-i\phi_L} & \Omega e^{-i\phi_L} & 0 \\ \Omega e^{i\phi_L} & 0 & 0 & \Omega e^{-i\phi_L} \\ \Omega e^{i\phi_L} & 0 & 0 & \Omega e^{-i\phi_L} \\ 0 & \Omega e^{i\phi_L} & \Omega e^{i\phi_L} & -2\Delta \end{pmatrix}.$$

5.2.3 Rydberg states

Specifically, for the milestone, we will look at the case where the qubits are formed by atoms and $|0\rangle$ and $|1\rangle$ are two atomic states. We want these atoms to be able to interact with each other. While the Hamiltonian of Eq. (5.4) can be used for describing two atoms interacting with an EM field, it does not include an interaction *between* the atoms. We therefore need to modify this Hamiltonian judiciously.

As atoms are neutral, they do not interact sufficiently strongly to create quantum entanglement. One solution to this problem is to excite the valence electron to a state with a high principal quantum number, $n > 50$ say. These highly-excited states are known as Rydberg states and readily interact with each other through the van der Waals interaction — an electrostatic interaction depending on the charge distribution of the respective atomic states and shifting their energy. If we take one state $|0\rangle$ to be the ground state or a low excited state and $|1\rangle$ to be a Rydberg state $|{\rm r}\rangle$, then the doubly excited state $|{\rm rr}\rangle$ experiences a van der Waals energy shift of $\hbar V = C_6/R^6$, where R is the distance between the two atoms (i.e., the internuclear distance) and C_6 is a constant increasing very rapidly with the principal quantum number. In this case the two-atom Hamiltonian in the basis $|00\rangle$, $|0{\rm r}\rangle$, $|{\rm r}0\rangle$ and $|{\rm rr}\rangle$ has the form

$$\mathcal{H} = \frac{\hbar}{2}\begin{pmatrix} 2\Delta & \Omega e^{-i\phi_L} & \Omega e^{-i\phi_L} & 0 \\ \Omega e^{i\phi_L} & 0 & 0 & \Omega e^{-i\phi_L} \\ \Omega e^{i\phi_L} & 0 & 0 & \Omega e^{-i\phi_L} \\ 0 & \Omega e^{i\phi_L} & \Omega e^{i\phi_L} & -2\Delta + 2V \end{pmatrix}. \tag{5.4}$$

We shall use this form of the Hamiltonian in the milestone project. We note, here, that $\mathcal{H}$ is time-independent if Δ, Ω, ϕ_L and V are all constant. The milestone will involve considering this case, but also the case a time-dependent detuning. Varying other parameters could also be of interest. For example, if we make Ω a function of time we can explore the effect of the temporal profile of an EM pulse.

5.3 Milestone project

The aim of the milestone is to solve the Schrödinger equation

$$\mathrm{i}\hbar\partial_t|\Psi(t)\rangle \quad = \quad \mathcal{H}|\Psi(t)\rangle \tag{5.5}$$

for the two-atom Hamiltonian given by Eq. (5.4), given an initial state vector $|\Psi(0)\rangle$ ($|\Psi(0)\rangle = |\Psi(t)\rangle$ at $t = 0$). This equation will first need to be solved for the case where the Hamiltonian is time-independent and then for the case where parameters in the Hamiltonian vary with time.

5.4 Work plan

5.4.1 Design of your program

The inputs into your programme include the Hamiltonian which depends on the parameters Ω, Δ, ϕ_{L} and V, and an initial state vector.

There is a simple method to solve Eq. (5.5) for a time independent Hamiltonian: Find the eigenvalues and eigenvectors of $\mathcal{H}$ and rewrite the initial state in terms of these eigenvectors (use a rotation matrix R made up from the eigenvectors to this effect). Evolving each eigenvector in time simply amounts to multiplying it by $\mathrm{e}^{-\mathrm{i}E_j t/\hbar}$, where E_j is the corresponding eigenvalue (i.e., the corresponding eigenenergy of $\mathcal{H}$, not an eigenenergy of $\mathcal{H}_0$ as in Section 5.2.1). Finally, after time evolution, transform back to the starting basis (the $|00\rangle$, $|0\mathrm{r}\rangle$, $|\mathrm{r}0\rangle$, $|\mathrm{rr}\rangle$ basis). In brief, the solution to Eq. (5.5) is then obtained as

$$|\Psi(t)\rangle \quad = \quad \mathsf{R}^{\dagger}\mathsf{D}(t)\mathsf{R}|\Psi(0)\rangle \ , \tag{5.6}$$

where $\mathsf{D}(t)$ is the diagonal matrix whose diagonal elements are the exponentials $\mathrm{e}^{-\mathrm{i}E_j t/\hbar}$. This method is largely the same as that leading to Eq. (5.2) in the case of $\mathcal{H}_0$, except that here $\Psi(0)\rangle$ and $\Psi(t)\rangle$ are not written in a basis of eigenvectors of the Hamiltonian.

However this method no longer works if the Hamiltonian is time dependent. Instead, for this case, use an iterative method known as the Trotter-Suzuki decomposition. Namely, divide the time interval into small time steps of length δt and propagate the solution over each time step using the formula

$$|\Psi(t+\delta t)\rangle \quad = \quad \mathrm{e}^{-\mathrm{i}\mathcal{H}(t)\delta t/\hbar}|\Psi(t)\rangle \ , \tag{5.7}$$

starting with $|\Psi(0)\rangle$ at $t = 0$. This latter approach can be justified as follows. If δt is sufficiently small, then $\mathcal{H}$ can be taken to be effectively constant during each of these time intervals. Within this approximation, Eq. (5.7) gives the correct evolution of the state vector over each time step. For the Hamiltonian considered, the result obtained in this way converges towards the correct solution in the limit where δt is reduced to zero. (If you took Theoretical Physics 2 last academic year you may recognize that Eq. (5.7) expresses the time-evolution of $|\Psi(t)\rangle$ in terms of the evolution operator, i.e., as $|\Psi(t+\delta t)\rangle = \mathcal{U}(t+\delta t, t)|\Psi(t)\rangle$, under the assumption that $\mathcal{H}(t)$ is constant between t and $t+\delta t$; however, you won't need to know this formalism for this project).

These are not the only ways of solving Eq. (5.5) for the present Hamiltonian. In particular, this equation could be solved as a system of ordinary differential equations using standard numerical methods. However, for the milestone you are asked to use the methods outlined above.

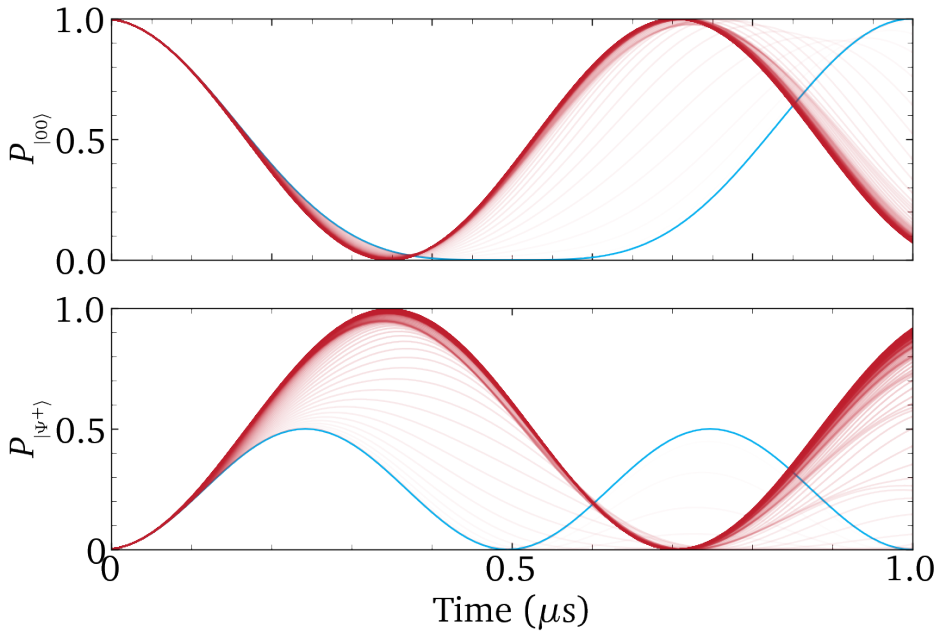


Figure 5.1: The time dependence of a 2-atom system with resonant excitation to the Rydberg state. We plot the probability to be in the state $|00\rangle$ and the maximally entangled Bell state $|\Psi^+\rangle = \frac{1}{\sqrt{2}}(|0\mathrm{r}\rangle + |\mathrm{r}0\rangle)$. The blue line shows the case of $V = 0$. The Rabi frequency is given by $\Omega/(2\pi) = 1$ MHz. The red lines show $V/(2\pi)$ in the range 0 to 100 MHz.

5.4.2 The milestone problem

As already mentioned, you should solve Eq. (5.5) using the two methods described above. Start with the time-independent case, for which you should use the eigenenergies method. Your program should be able to reproduce the plot shown in Fig. 5.1. Investigate the residuals with respect to analytical solutions for the cases $V \approx 0$ and $|V| \gg |\Omega|$.

Next, solve Eq. (5.5) for a time dependent detuning, keeping the Rabi frequency constant, using the Trotter-Suzuki method. Specifically, your program should be able to reproduce the results shown in Fig. 5.2. This example shows how one can produce the Bell state $|\Psi^+\rangle$ with high fidelity by sweeping the detuning through resonance. Investigate the effect of using different sweep profiles, e.g. linear vs. cubic in time, and how large a time step can be used for an accuracy of 4 significant figures.

5.4.3 Extensions

The Extension is flexible but as a minimum it should relate to quantum simulation or quantum computation. Some suggestions are given below.

- **Two-qubit gates**: In a Rydberg quantum computer the most basic entangling gate is the CZ, where the $|11\rangle$ state accumulates an additional phase of π relative to the other terms. To realise this we consider a qutrit with states $|0\rangle$, $|1\rangle$ and $|\mathrm{r}\rangle$. For two qutrits, when we drive $|11\rangle$ to $|\mathrm{rr}\rangle$ the interaction gives rise to a different phase evolution. The aim is to engineer the parameters such that the complete time evolution in Eq. (5.7) (or equivalent if you'd use another numerical method) has

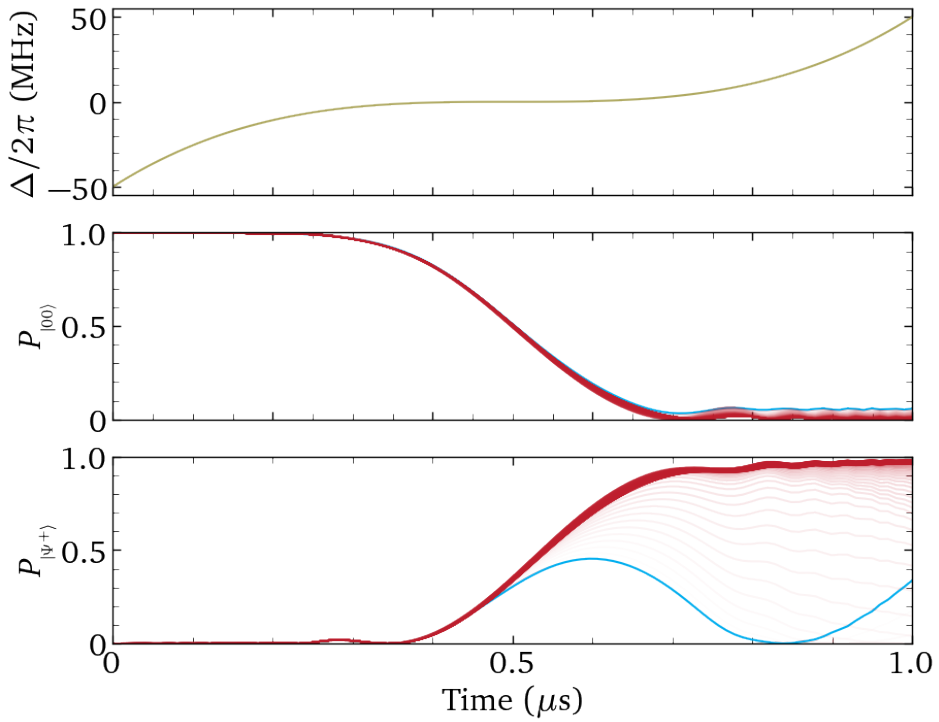


Figure 5.2: The time dependence of a 2-atom system as the detuning is swept through resonance, $\Delta(t) = 8\Delta_{\max}[(t - t_{\max}/2)/t_{\max}]^3$ with $\Delta_{\max}/(2\pi) = 50$ MHz and $t_{\max} = 1$ μs. The other parameters are the same as in Fig. 5.1.

the form of a CZ gate, i.e., a gate evolving a pair of atoms initially in the state

$$a_{00}|00\rangle + a_{01}|01\rangle + a_{10}|10\rangle + a_{11}|11\rangle$$

into the state

$$a_{00}|00\rangle + a_{01}|01\rangle + a_{10}|10\rangle - a_{11}|11\rangle \ .$$

See e.g. Ref. [1] for theory and [2] for experiments.

- **Many qubits**: The simulation in Fig. 2 can be reproduced for N qubits. How does the computation time scale with N? Is there anything you can do about it? For many atoms in a linear array, the output depends on the spacing between the atoms, R, the Rabi frquency Ω and the interaction strength, C_6, see e.g. Fig. 2 in Ref. [3].

- **Qutrits and qudits**: In many cases we need more that two quantum states to model a quantum process. Look into what has recently been done on that front, and investigate an interesting case numerically.

- **Decoherence**: Excited quantum states have a finite lifetime. Also the fields used to drive qubits are noisy. We can model decay and decoherence using the optical Bloch equations which described the time evolution of the density matrix. Write an OBE simulation to reproduce Fig. 2 with the addition of decay.

References

1. *Time-Optimal Two- and Three-Qubit Gates for Rydberg Atoms*, S Jandura and G Pupillo, Quantum **6**, 712 (2022);
https://doi.org/10.22331/q-2022-05-13-712.

2. *High-fidelity parallel entangling gates on a neutral-atom quantum computer*, SJ Evered *et al*, Nature **622**, 268 (2023);
https://www.nature.com/articles/s41586-023-06481-y.

3. *Probing many-body dynamics on a 51-atom quantum simulator*, H Bernien *et al*, Nature **551**, 579 (2017);
https://www.nature.com/articles/nature24622.